TikTok KOL Matcher

Setup instructions. Source code: https://github.com/Jamessurapat26/tiktok-kol-matcher

Option	Use it to	Needs
1. Demo with Docker (start here)	See the whole product on sample data	Docker with Compose. No keys and no cost
2. Live mode	Match a real brand with real TikTok data	Docker and API keys. Costs money per job
3. Without Docker	Develop, run tests, or use the CLI	Python 3.13 with uv, Node 24, pnpm 11

1. Demo with Docker

Demo mode runs the full pipeline on built-in sample data. It needs no `.env` file and makes no paid calls.

```
git clone https://github.com/Jamessurapat26/tiktok-kol-matcher.git
cd tiktok-kol-matcher
docker compose up --build
```

The images take about 2.2 GB of disk, so the first build can take several minutes. When it finishes:

1. Open **http://localhost:5173**.
2. Press the ลองดูตัวอย่าง button ("try an example") to fill in the sample brand, then submit the form.
3. Check the result. The job finishes in a few seconds, and the top creator should be **@fpscoach.oat**.

Sample data: the brand and every creator in demo mode are made up, so their TikTok links may not open. The web app is in Thai because it is written for Thai marketers.

Press Ctrl-C to stop. Run `docker compose down` to remove the containers, or `docker compose down -v` to delete saved jobs and caches as well.

2. Live mode

Live mode reads a real website and Facebook page, searches TikTok through Apify, and uses a real LLM. Create the settings file from the template:

```
cp .env.example .env
```

Then fill in these values in `.env` and leave the rest blank:

```
APIFY_TOKEN=your-apify-token
LLM_BASE_URL=https://openrouter.ai/api/v1
LLM_MODEL=deepseek/deepseek-v4.1-flash-1017
LLM_API_KEY=your-openrouter-key
JINA_API_KEY=your-jina-key
```

Variable	Used for	Where to get it	If it is missing
`APIFY_TOKEN`	Facebook page, TikTok search, creator profiles	Apify Console, under API & Integrations (console.apify.com)	The app stays in demo mode
`LLM_API_KEY`	Brand brief, search queries, creator judging	OpenRouter (openrouter.ai/keys)	LLM steps use mock answers
`JINA_API_KEY`	Web research on the brand	jina.ai	Web research is off. The website reader still works at a lower rate limit

This is the project's live LLM setup. If `LLM_BASE_URL` and `LLM_MODEL` are left blank, the code uses NVIDIA Nemotron on NVIDIA's API instead, with a key from build.nvidia.com. Any OpenAI-compatible provider works the same way.

Start the app, and in a second terminal confirm that every service is live:

```
docker compose up --build

# in a second terminal
curl -s http://localhost:8000/api/health
```

The reply must show `"mode":"live"` and every value under `"defaults"` as `true`. A value of `false` means that service will use mock data, so a job can succeed with results that are not fully real. Keys appear only as true or false, never as their values.

Now enter a real brand's website and Facebook page. The sample brand exists only in demo mode and fails in live mode.

One live job	What to expect
Run time	About 7-11 minutes in recorded runs
Apify cost	Estimated at up to $1.55, with a check before each run. Set `APIFY_JOB_BUDGET_USD` to change it
LLM cost	Separate, depends on your provider, and not tracked by the app
First run	Also downloads the embedding model into the Docker volume

Do not commit `.env`. It is already in `.gitignore`. The API has no login, so Compose binds both ports to localhost only. Do not expose them publicly.

3. Without Docker

Install Python 3.13 with uv, Node 24, and pnpm 11. Then, from the repository root:

```
./start.sh --demo     # demo mode: API on :8000, web app on :5173
./start.sh            # live mode, using the keys in .env
```

Live mode also needs the local embedding packages. Install them once before the first run:

```
cd backend && uv sync --extra embed && cd ..
```

Command-line version

The CLI runs the same pipeline and prints the ranked table. Without any `--live-*` flag it uses mocks only:

```
cd backend
uv run python -m app.cli match \
  --website https://example-brand.co.th \
  --facebook https://www.facebook.com/example-brand \
  --top-k 10
```

Run the tests

The tests make no network calls and need no keys.

```
cd backend && uv run pytest            # 510 tests
cd frontend && pnpm install && pnpm lint && pnpm test   # 84 tests
```

Troubleshooting

Problem	Cause and fix
`ports are not available` or `Port 8000 is already in use`	Another app is using port 8000 or 5173. Stop it, then start again. Check that it is not an older copy of this app, which would answer your requests instead
Health shows `"mode":"demo"` after adding keys	`APIFY_TOKEN` is blank, or the containers kept their old settings. Run `docker compose down`, then `docker compose up`. Also check that `DEMO_MODE` is not set to 1
A value under `"defaults"` is false	That service's key is missing. Without Docker, `live_embed` is false until you install the packages in section 3
Job fails with `401 Missing Authentication header`	The LLM key is missing, wrong, or belongs to a different provider than `LLM_BASE_URL`
Job fails with `no usable brand source`	Neither the website nor the Facebook page could be read. Check both URLs, and that the Facebook page is public
`Apify rejected the token (HTTP 403)`	The Apify token is invalid, or the account cannot run the actor. Check the token and the account's credit

The full variable list is in `.env.example` and the README's Configuration section. The HTTP API is described in `docs/reference.md`.